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# Coherent Terahertz Smith–Purcell radiation from a two-section model

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ABSTRACT

This paper presents a two-section model to produce beam bunching and obtain coherent Terahertz (THz) Smith–Purcell (SP) radiation. Based on the mechanism of the Cherenkov oscillator, a continuous beam is bunched in the first section without external signal. In the second section, the coherent THz-SP radiation is stimulated by the bunched beam interacting with open grating. The Particle-In-Cell (PIC) simulations show that the beam is effectively bunched and the strongest radiation is observed at 51° with the frequency of 194 GHz in this scheme.

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## 1. Introduction

The Terahertz (THz) sources, a currently active research area, are of importance in a variety of applications to biophysics, medical, and materials science [1,2]. The coherent Smith–Purcell (SP) radiation is a promising alternative in the development of a compact, tunable, and high-power THz device. However, it is necessary to improve the performances of such kinds of devices.

Recently, several articles [3–16] have discussed the SP radiation mechanism and characteristics. They concluded that high-quality bunches are the key to generating coherent SP radiation.

We have previously presented a study of obtaining coherent THz-SP radiation from a two-section grating system [17]. By means of an input signal the electron beam is bunched in traveling wave domain in the first section of the model. To obtain a compact and relatively inexpensive coherent THz source, we will explore in this article how to generate a train of periodic electron bunches mainly based on the mechanism of the Cherenkov oscillator.

This paper is organized as follows: Section 2 contains a description of the physical model. In Section 3, the simulations are carried out using a CHIPIC [18] code, and we analyze the physical process of generating THz radiation. The conclusions are given in Section 4.

## 2. The physical model

A schematic diagram of the two-section model is given in Fig. 1. The system consists of a grating with a flat conducting roof and an open grating .The main parameters of the grating and the electron beam are summarized in Table 1.

The dispersion curve of the first section [19] is shown in Fig. 2. The operating point P of Fig. 2 is indicated with $f = 97$ GHz, i.e., the intersection of the dispersion relation with the beam line for 40 keV electrons. Obviously, the first section operates like a diffraction oscillator, then the beam can be bunched without an input signal.

In the second section, the coherent SP radiation is stimulated by the bunched beam. Frequency of the SP radiation [20] observed at an angle $\theta$ as shown in Fig. 1 is

$$f = |n|\frac{c}{d}\left(\frac{1}{\beta} - \sin\theta\right)^{-1} \tag{1}$$

where $f$ is the frequency of the radiation, $n$ is the order of the radiation, $d$ is the grating period, $\beta c$ is the electron velocity, and $c$ is the speed of light.

## 3. PIC simulations

### 3.1. Descriptions of the simulation geometry

The simulations are carried out using a CHIPIC code. It is a finite-difference, time-domain code designed to simulate plasma Physics processes. The geometry for simulations is given in Fig. 3.

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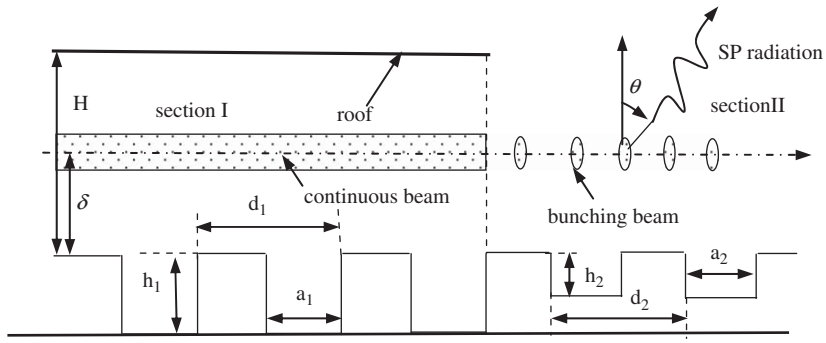


**Fig. 1.** Schematic illustration of the physical model.

**Table 1**
Main parameters for the two-section model.

| Parameters | First section | Second section |
|---|---|---|
| Beam energy | 40 keV | – |
| Current density | 200 A/m | – |
| Beam thickness | 0.4 mm | – |
| Beam-grating distance | $\delta = 0.3$ mm | – |
| Grating period | $d_1 = 0.5$ mm | $d_2 = 0.8$ mm |
| Grating groove depth | $h_1 = 0.625$ mm | $h_2 = 0.24$ mm |
| Grating groove width | $a_1 = 0.25$ mm | $a_2 = 0.24$ mm |
| Number of period | $N_1 = 89$ | $N_2 = 12$ |
| Grating roof height | $H = 0.75$mm | – |
| External magnetic field | 2 T | – |

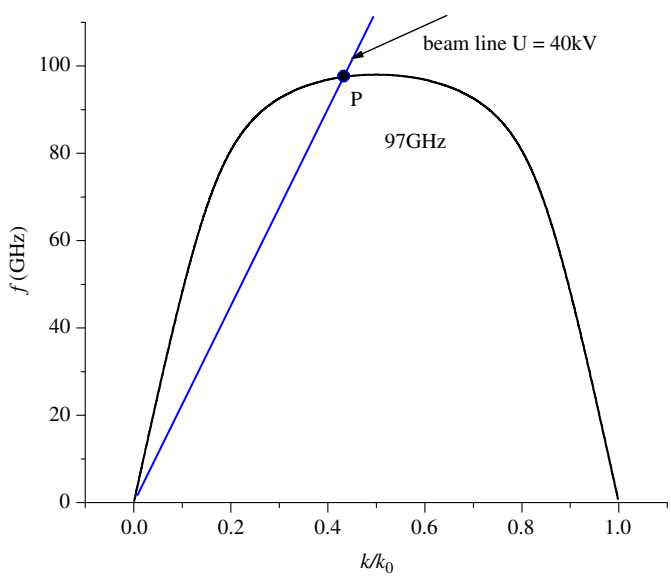


**Fig. 2.** Dispersion curve of the grating with a flat conducting roof. Parameters of grating: period $d_1 = 0.5$ mm, slot width $a_1 = 0.25$ mm, slot depth $h_1 = 0.625$ mm.

The roof and the grating are assumed to be perfect conductors; the latter has uniform rectangular grooves along the $z$ direction. A sheet electron beam propagates along the $x$ direction. It is a perfect beam produced from a small cathode located at the left boundary of the simulation area. The beam–wave interaction and radiation propagation occur in the vacuum box. The boundary is enclosed with absorbers. At the end of the first section, there is an attenuator, which prevents most of the electromagnetic waves from reaching the second section.

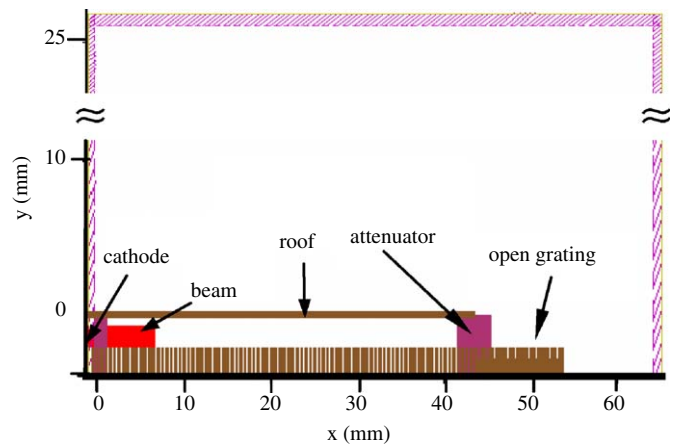


**Fig. 3.** Simulation geometry.

The simulation parameters of the grating and the electron beams are shown in Table 1. The parameters of the first section grating are the same as those in the previous article [17]. Since it is a two-dimensional simulation, it assumes that all fields and currents are independent of the $z$ direction. The current density is 200 A per meter. To operate the first section as an oscillator, the beam voltage is 40 kV.

### 3.2. Generation of periodic bunches

To illustrate beam bunching, we show the phase–space plots and energy modulation of electrons at time 3 ns, as shown in Fig. 4. The Particle-In-Cell (PIC) simulations show that bunching can be clearly observed as functions of both space and time. In Fig. 4(a) there is no evidence of the beam bunching above the first grating. But as soon as the beam reaches the second grating, strong bunching is visible. It would be because the groove depth ($h_2/d_2 = 0.3$) in the second grating is relatively low. The interaction between the beam and the slow wave is weak-coupling, which resembles a drift section to some extent. When the beam-by-velocity modulation in the first grating passes to the second grating, obvious bunching can be observed. From Fig. 4(b) we note that appreciable energy modulation is visible and the mean beam energy reduction is about 2 keV due to the beam–wave interaction. During the process of beam bunching, a portion of the beam energy reduction is transferred to the slow wave through a box whose height is 0.75 mm. The slow wave will radiate at the ends of the grating.

The modulation of the current is displayed in Fig. 5. Fig. 5(a) shows that bunching of the current reaches its maximum at the center of open grating. In Fig. 5(b), one clearly sees the bunching

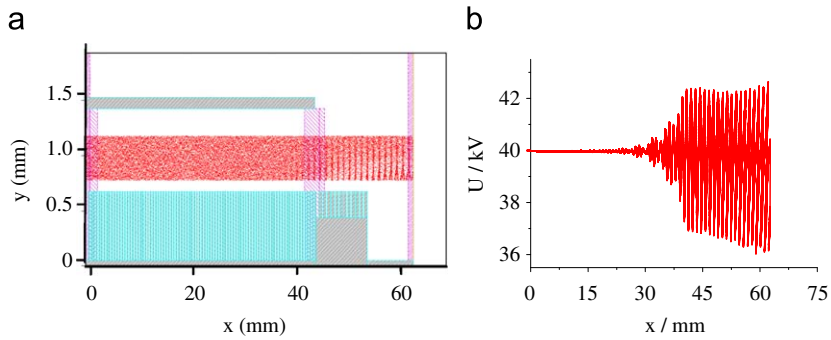


**Fig. 4.** Phase–space distribution: (a) density of electrons in the $x$–$y$ plane at 3 ns, bunching is evident, and (b) kinetic energy as a function of $x$ direction at the same time.

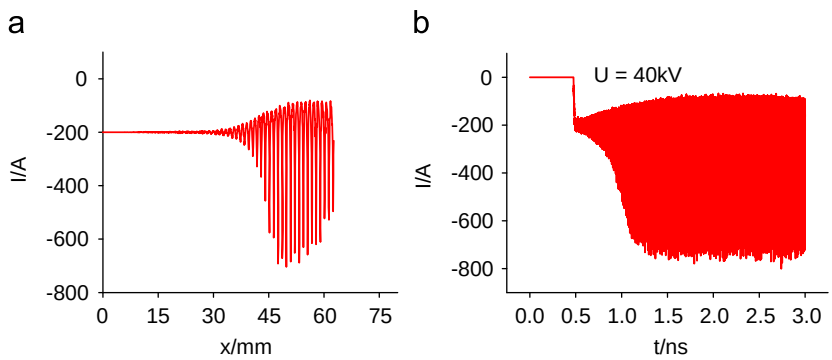


**Fig. 5.** Current evolution as functions of both space and time: (a) current as a function of $x$ direction at 3 ns, and (b) current as a function of time at the center of open grating.

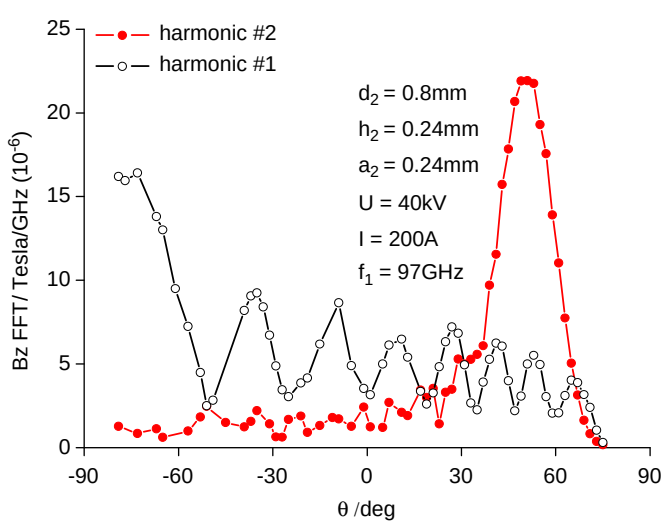


**Fig. 6.** The peak of FFT amplitude of $B_z$ as a function of angle, detected at the distance 21.25 mm from the open-grating center.

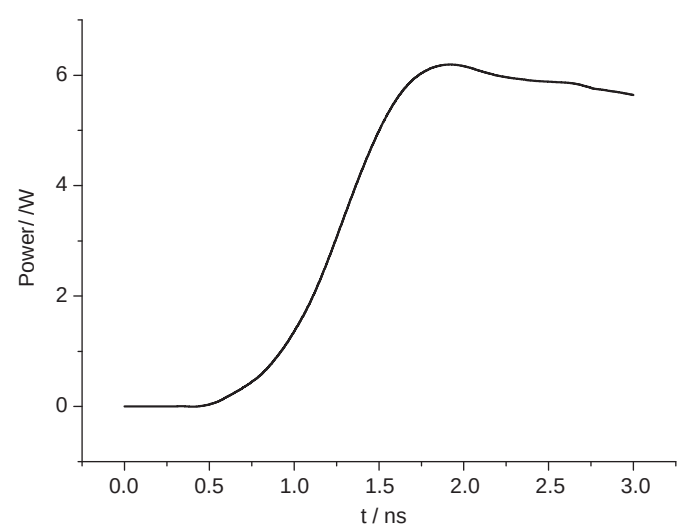


**Fig. 7.** The SP radiated power observed at the angle of 50–52°.

current is stable in time after about 1.3 ns. The simulation results show that a continuous beam is effectively bunched by the first section.

### 3.3. Coherent THz radiation

Here we focus on the radiation from the bunched beam interacting with the open grating. We observe the coherent SP radiation emitted at harmonics of the bunching frequency of 97 GHz. We use a grating of period 0.8 mm, which has 12 periods. The lowest SP frequency for 40 keV is 102 GHz. The $B_z$ detector is placed at the distance 21.25 mm from the center of the open grating. Because only the second harmonic radiation frequency of 194 GHz is allowed SP frequency of the first order ($|n| = 1$), we observe the distributions of the evanescent and the SP radiation waves, as illustrated in Fig. 6. From the Fast-Fourier-Transform (FFT) amplitude observed by the $B_z$ detectors, we know that the dominant radiation is the second harmonic of the bunching frequency, which peaks at an angle of about 51° corresponding to the SP radiation angle predicted by Eq. (1). Of course, due to the beam–wave interaction, which reduces the value of the particle

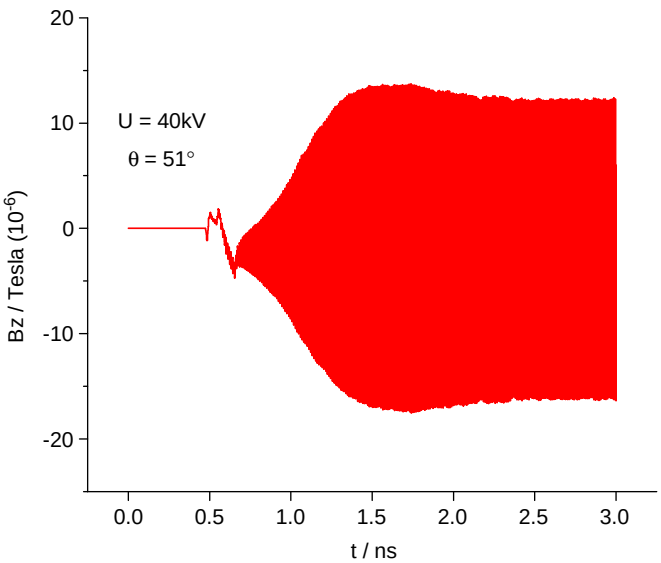


**Fig. 8.** Time signal of $B_z$ observed at point 21.25 mm and 51° from the open-grating center.

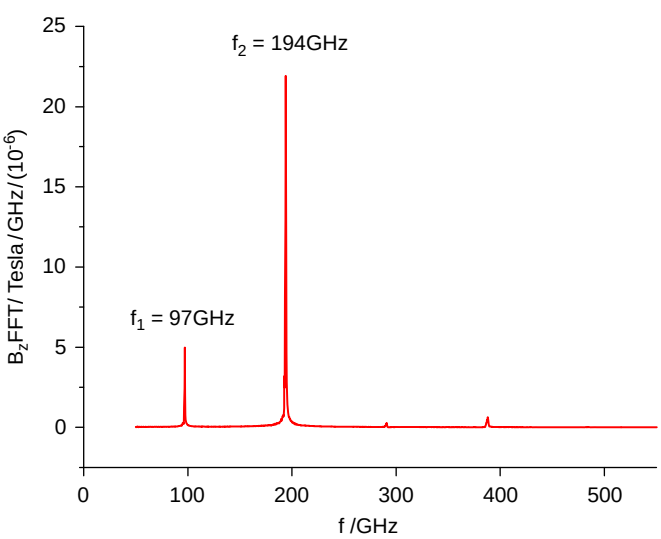


**Fig. 9.** FFT of time signal corresponding to Fig. 8.

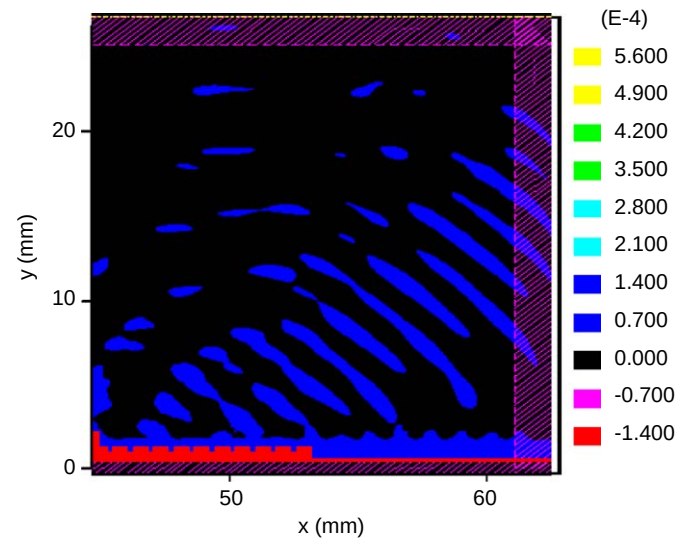


**Fig. 10.** Contour map of $B_z$, $t = 2.6$ ns.

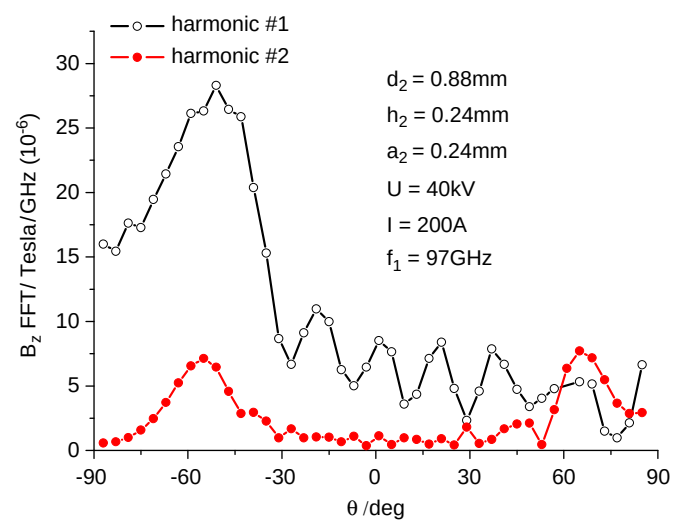


**Fig. 11.** The peak of FFT amplitude of $B_z$ as a function of angle for a grating of period 0.88 mm, detected at the distance 21.25 mm from the open-grating center.

velocity, there is a slight discrepancy between the simulation data for the radiation angle of 51° and the theoretical value 48° of the basic SP equation. According to Fig. 6, we see that the evanescent wave radiation reaches maximum at an angle of about −90° corresponding to the left end of the open grating. In Fig. 7, the simulation results show that the radiated power observed at the angle of 50–52° corresponding to the strongest radiation regime is about 5.6 W.

In Fig. 8 we show evolution of magnetic field $B_z(t)$ at an angle 51°, and the corresponding FFT shown in Fig. 9. The time signal shows rapid growth after 0.7 ns. From the contour plot of Fig. 10, one can easily understand that the dominant second harmonic radiates at the angle of about 51°, in agreement with what is shown in Fig. 6.

We also simulate a second grating of period 0.88 mm and the result is displayed in Fig. 11. The lowest SP radiation frequency is 92.8 GHz for period 0.88 mm and beam energy 40 keV, so the coherent SP radiation of the fundamental bunching frequency of 97 GHz is allowed. According to Fig. 11, the SP radiation contains two backward peaks and a small forward peak. The first backward peak (near −51°) in the first harmonic is the first order ($|n| = 1$ in Eq. (1)). The second peak (near −55°) is the second order ($|n| = 2$ in Eq. (1)) for the second harmonic, which should occur at the same theoretical angle −57° predicted by Eq. (1) as the first-order peak in the first harmonic. There is a slight discrepancy between the simulation data and the theoretical value. The small forward peak (near 65°) in the second harmonic is the first order ($|n| = 1$ in Eq. (1)). The first-harmonic peak of the coherent SP radiation is much stronger than the second harmonic.

## 4. Conclusions

In this paper, we have presented a study of obtaining coherent THz-SP radiation from a two-section grating system. Based on mechanism of the Cherenkov oscillator, a continuous beam is effectively bunched through the first section. In the second section, the coherent SP radiation has been produced by the bunched electrons interacting with the open grating. The strongest radiation is observed at 51° and the radiation frequency

is 194 GHz at parameters used in the simulation. The coherent THz-SP radiations are emitted at frequencies that are integer multiples of the bunching wave frequency, and at the corresponding SP angles. The results show that the process is stable in time and along the grating. The coherent SP radiation will occur in the other frequency band when proper parameters are chosen. In this scheme, the periodic bunches are generated spontaneously, with no need for an input signal, thereby reducing the cost of the source.

## Acknowledgement

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